LESSON PLAN

BASIC RADIO & AVIONICS (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose and principle operation of aircraft pressure gauges, pilot static system, altimeter, ASI, and VSI.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose and principle operation of aircraft pressure gauges, pilot static system, altimeter, ASI, and VSI. Practical on Pitot & Static instruments. Safety precautions while working on system
5. Specific Objectives	25.1.1.1 Trainee will be able to understand basic instrument gauges, instrument type auto pilot and maintenance.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on Pitot & Static instruments; Safety precautions while working on system; Instructor supervision during handling
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the static system?
(b) Name three instruments powered by the static system.
(c) Why is static pressure accuracy critical for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Purpose of aircraft pressure gauges • Components of the pitot-static system • Principle of operation for Altimeter • Principle of operation for ASI • Principle of operation for VSI	L & Ds	PPT, IB	15 min
(b)	• Static source location and routing • Impact of blockage on instrument readings • Maintenance and inspection procedures • Troubleshooting common static system faults • Calibration and testing protocols	L & Ds	PPT, PDF	20 min
(c)	• Safety precautions for static system work • Handling Pitot tubes and static ports • Use of protective equipment • Lockout/tagout procedures • Post-maintenance verification steps	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a change in static pressure affects the altimeter reading.
(b) Describe the safety steps required before inspecting a static port.
(c) Differentiate between the function of the ASI and the Altimeter.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between the static systems used in LP 02 and the upcoming LP 03 instruments.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Capacitor type fuel quantity indicator, angle of attack indicator, gyroscopic instruments attitude indicator, turn and slip indicator, direction indicator. Practical on various instruments.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. The master syllabus says 'pilot static system'; confirm whether this is a typo for 'pitot-static system'.
2. Confirm the master syllabus reference book/BR for this subject.
3. Confirm the class existing knowledge and prerequisites.
4. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
5. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
6. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min